MIND: Mental Health Impact and Navigation due to Digital Technology

Mir Faiyaz Hossain Shajreen Tabassum Diya Fariha Moazzema Raisa

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Abstract

About 1 in every 8 people suffer from mental health issues worldwide [20]. Mental wellbeing is a serious matter of concern as poor mental health can hinder a person's quality of life. Along with this, the increase in technology usage also contributes to deteriorating our mental wellbeing. Our work aims to predict the impact of technology on an individual's mental health.To proceed with the work, after doing in-depth research about what could be an ideal set of questions, we initiated preparing a questionnaire for collecting data. The questionnaire had 5 sections that comprised 38 questions in total that asked questions about the participants, participants' interaction with technology and how they cope stress. The questionnaire was filled by 624 participants but after checking carefully 613 data were finally considered. We used nine classifiers, one feature selection and three hyperparameter tuning techniques. The maximum achieved accuracy was 72% for logistic regression, and recall was 67% for Random Forest.In the end, an explainable AI framework "LIME" was used to get a thorough understanding of how the machine learning model derived its predictions.

1 Introduction

In this fast-paced age of the internet, the usage of digital technology is no longer constrained to professionals and technocrats [19]. Digital devices have become a part of everyday life. Digital technology has become a medium of communication, source of information and entertainment. With more and more intensive use of technology, there are evidences of positive and negative impacts of technology on our mental health. There are rising concerns in the context of negative impacts and addictive behaviors towards technology usage. Internet Addiction and Smartphone addiction are termed as Problematic Internet Usage (PIU) and Problematic Smartphone Usage (PSU) [2]. According to a survey conducted on students at Kerman University of Medical Sciences, where 920 students participated in the survey, 21% of the participants were addicted to technology usage and of them around 22.2% of the students spent more than 15 hours per week on average on the internet [10].

To further investigate how technology usage affects mental health, we devised a questionnaire of 38 questions related to tech-usage and carefully picked relevant fields related to technology, to understand how usage affects mental health. There were 624 participants in our survey, and we used various Machine Learning techniques to assess the mental wellbeing of an individual through the data collect. Our main goal was to find how different aspects of technology usage affects mental health. The major contributions of our work are listed below:

- We devised a questionnaire of 38 questions from different domains associated with an individual and technology usage, and collected data through Google Forms, and gathered 624 responses. The highlight of our questions were the inclusion of the ability of an individual to practice ”Digital Detox”.
- We used various machine learning algorithms to classify the mental state of an individual into three classes, using our carefully selected features.
- We applied feature selection and hyperparameter tuning techniques to increase the performance of our models.
- We used LIME explainer, which is an explainable AI tool to understand how the predictions were being made.

The rest of the paper is structured according to the following order: In section 2, we have summarized our literature review, in section 3 we discussed the methodology of our work in details. In section 4, the result analysis of our work is given. Section 5 is the discussion of our work, section 6 and 7 shows the limitations and conclusion with future scope.

2 Literature Review

Mental health is a very important aspect to be taken into consideration for an individual's wellbeing. Until recently there have only been conventional approaches to dealing with mental health concerns but with the rise of machine learning techniques, there are works that cater to the need of mental wellbeing. This section discusses some of the relevant work done in assessing mental health and technology usage.

2.1 Related work on problematic technology usage

With an increase in the use of technology, there are many factors which can take a toll on an individual's mental health. Being actively connected to technology can also result in psychological issues. In [5], support vector machine (SVM) was used among Chinese college students for detection of internet addiction disorder. The data was collected using multiple personality questionnaires and the model showed accuracy of 96.32%. In [8], IAT data was used to determine the problematic use of the internet. Techniques that were applied in this paper included Logistic Regression, Naive Bayes and Random Forest. On the other hand, the model in [9] was a rule based model which had an extended classifier that combined reinforcement learning and Evolutionary Computation.

In [17] , they have collected a large amount of data related to smartphone usage. After that they used NaIve Bayes, AdaBoost and Support vector machine to identify problematic use of smartphones. Finally, they had an accuracy of 89.6%. In [16] problematic social media use (PSU) was modeled using Artificial Neural Network (ANN) and Support Vector Machine (SVM) and had 15 predictor variables to predict PSU. In PSU paper, data was collected from 1097 participants to analyze symptoms and characteristics like demographics, psychological measures of PSU, depression and anxiety symptoms, etc. Different kinds of machine learning algorithms were used to train the model.

2.2 Related works in mental health assessment

In [18], Bangladeshi students' data was collected using a questionnaire which had 106 questions and 684 respondents. The model used five hyperparameter tuning and nine feature selection methods to boost the performance of predicting depression and that led to accuracies of 98.08%, 94.23%, and 92.31% were obtained using Random Forest, Gradient Boosting, and CNN models, respectively. To predict and analyze depression and anxiety of undergraduate students [11] used ensemble based machine learning model. The model had a 0.67 Auc score on the validation set for the XGBoost model. In [22] Zulfiker and his team created a dataset by collecting data from Bangladeshi citizens and applied six different machine learning classifiers and three feature selection methods to predict depression and extract relevant features.The AdaBoost technique with the SelectKBest feature selection approach showed 92.6% classification accuracy.

In table 1, we have grouped and summarized our literature review.

Table 1: Summary of Papers and Machine Learning Models

S.I	Paper	Dataset	Machine Learning Models	Accuracy
1	[5]	Chinese college students, CIAS	Support Vector Machine	96.32%
2	[8]	IAT data	Logistic Regression, Naive Bayes, Random Forest	0.84 (ROC-AUC)
3	[9]	CIAS dataset	XCSR (technique)	100%
4	[17]	Data related to smartphone usage	Naive Bayes, AdaBoost, Support Vector Machine	89.6%
7	[18]	Bangladeshi students'	Random Forest, Gradient Boosting, CNN models	98.08, 94.23, 92.31
8	[11]	Undergraduate students	XGBoost	0.67 AUC score
9	[22]	Bangladeshi citizens	AdaBoost	92.60%

3 Proposed Work

In this section, a detailed description of the methodology applied is given. Figure 1 shows the life cycle of our research. This section includes the preparation of questionnaire, the dataset collection phase, the exploratory data analysis phase, the dataset cleaning and preprocessing techniques used, the classifiers used in training and evaluation, and metric optimization techniques applied in our work.

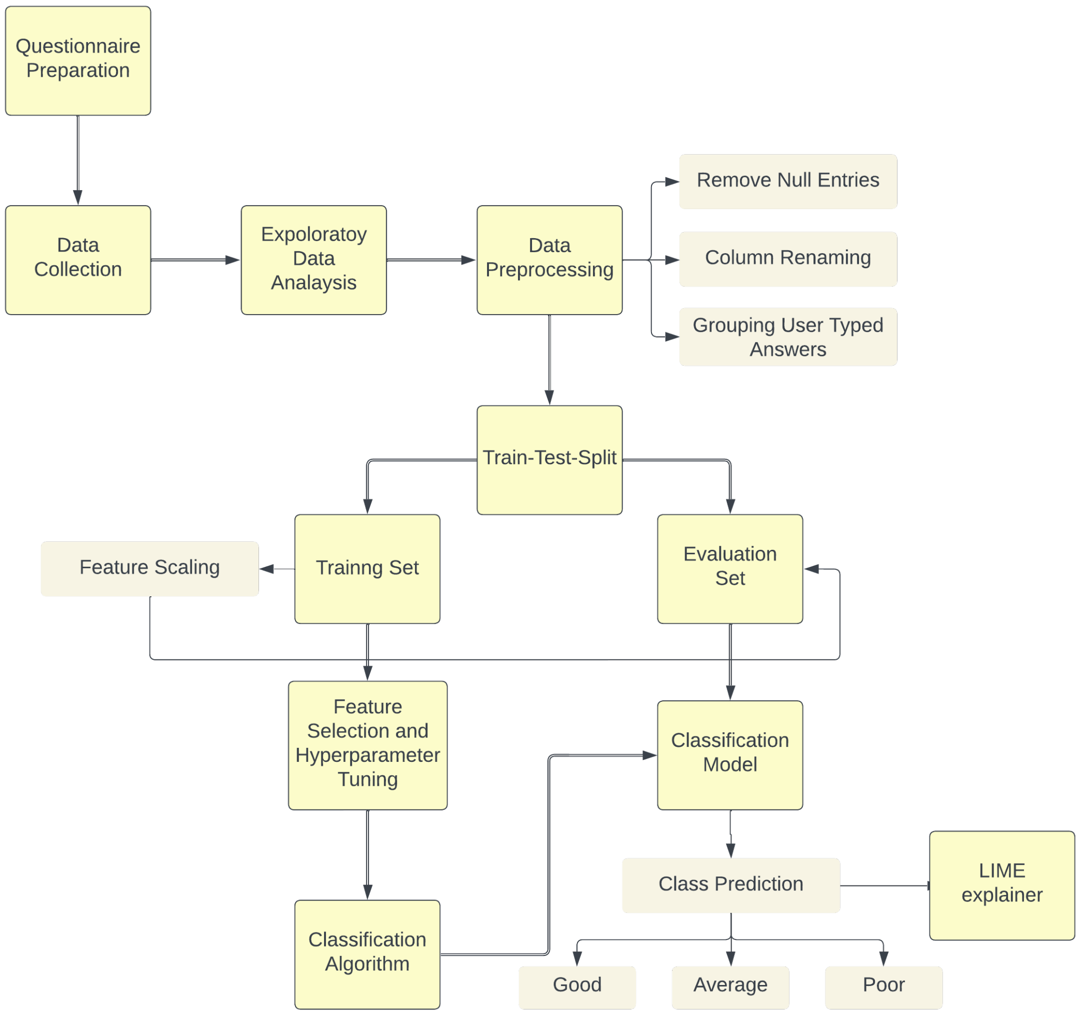


Figure 1: Methodology

3.1 Questionnaire Preparation

After a thorough literature review about the domain of our research, we devised a questionnaire consisting of 38 questions distributed among five different categories. The five categories were namely - demography, technology usage, mental health assessment, digital detox practices, and coping mechanisms. The demography section covered personal questions like age, gender and occupation. The technology usage section asked questions like type of content consumed, emotions associated with technology, purpose of using technology, frequency of usage, etc. Mental health assessment portion included questions to analyze the mental state and its symptoms like physical and emotional symptoms, isolation feeling and difficulty in focusing. Then we moved to ask questions to know if the participants practiced digital detox, what motivated them and what they felt about it. Lastly, we dived into the section of coping mechanisms where we asked for participants' coping strategies and its effectiveness. Figure 2 summarizes the content of the questionnaire.

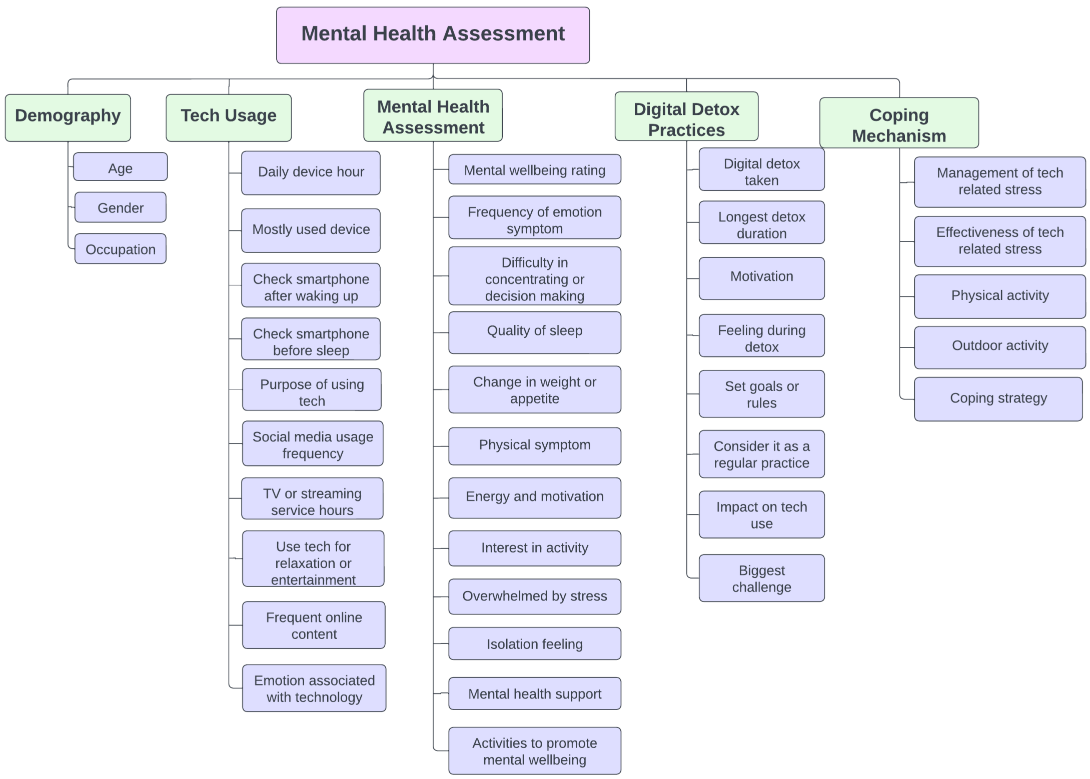


Figure 2: Questionnaire breakdown in five different categories

3.2 Dataset Collection

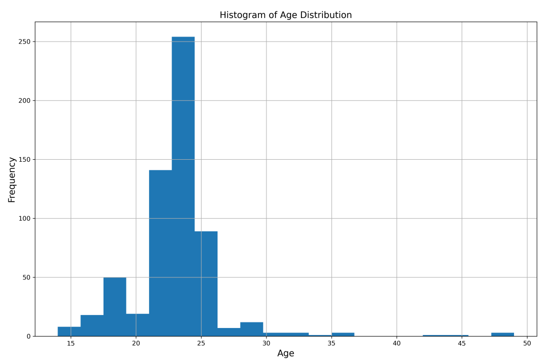
The dataset used in the training and evaluation process was collected using Google forms. Google forms is an online tool for creating surveys and collecting data. Google form was used to track responses, and visualize the distribution of answers using different statistical charts. After extensively collecting data for three weeks, we received responses from various age groups, but the majority of the responses were from people in their early twenties. We also received a good number of responses from teenagers, and few responses from people beyond the age of thirty, In total, there were **624 respondents** who participated in our survey. Each respondent agreed to sign a voluntary participation consent agreement. In this survey, we did not collect any personal information, and analyzed the data in an aggregated manner.

3.3 Exploratory Data Analysis

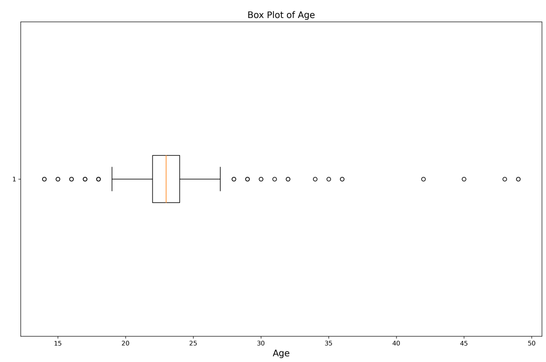
This section dives into the details of the dataset. We performed exploratory data analysis to get more information about our dataset. Pandas-profiling report showed that there were no duplicate or missing entries in our dataset. The dataset consist of thirty eight variables, and of them thirty six were categorical and two numerical.
The first numerical attribute in our dataset is **Age**. Figure 3 shows the representation of the data in histogram and box-plot. In Figure 3a, the histogram is positively skewed. The peak age in this histogram is 23.65. In Figure 3b, the lower-quartile (Q1) is 22.0, the median is 23.0 and the upper-quartile (Q3) is 24.0. The lower and upper whiskers are 19.0 and 27.0 respectively and are calculated using equations 1 and 2 :

$$\text{Lower whisker} = Q1 - 1.5 \times \text{IQR} \tag{1}$$

$$\text{Upper whisker} = Q3 + 1.5 \times \text{IQR} \tag{2}$$



(a) Age Distribution using Histogram



(b) Age Distribution using Box-plot

Figure 3: Age Distribution

To visualize the duration of time spent on digital devices by different age groups in a day, we constructed a box-plot. For Figure 4 on the next page, the summary statistics is tabulated in 2. The table shows that the median age for all the categories in daily usage of device in hours is 23.

Table 2: Summary Statistics for Age Across Daily Device Usage Hours

Daily Device Hour	Mean	Std.	1st Quartile	Median	3rd Quartile	Min	Max
Less than 1 hour	22.6	3.79	21.0	23.0	24.0	15.0	36.0
1-2 hours	22.9	3.46	22.0	23.0	24.0	14.0	48.0
3-4 hours	22.75	3.57	22.0	23.0	24.0	14.0	49.0
5-6 hours	22.2	3.11	22.0	23.0	24.0	17.0	25.0
More than 6 hours	23.0	3.6	22.0	23.0	24.0	14.0	49.0

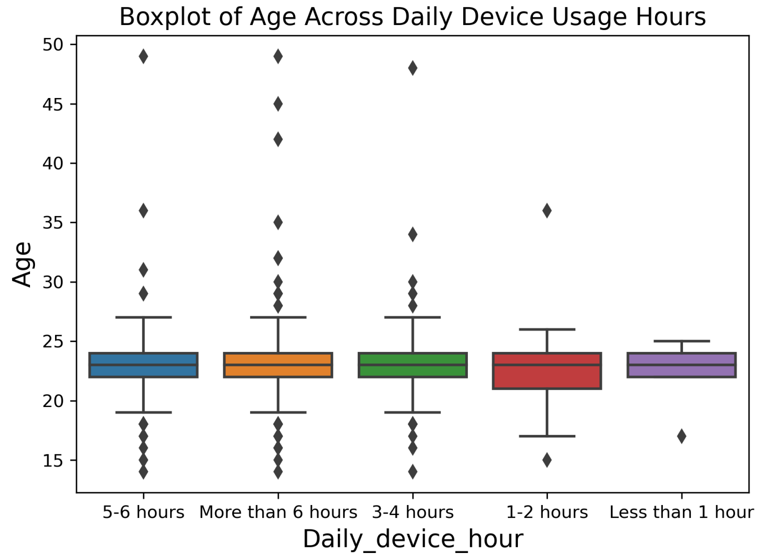


Figure 4: Box-plot of Age Across Daily Device Usage Hours

To represent the duration of hours spent on different digital devices, we have used a heatmap. In Figure 5, smartphone usage has the highest occurrence for all duration groups, whereas tablet usage has the least occurrence. Our dataset contained a total of 406 smartphone users, 201 Computer/Laptop users, 1 Tablet user and 5 users used other devices.

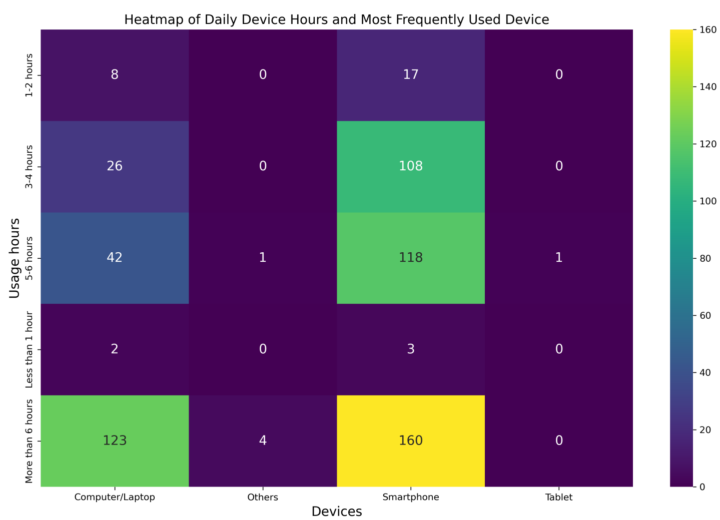


Figure 5: Heatmap of Daily Device Hours and Most Frequently Used Device

We have asked individuals to rate their mental wellbeing from 1-5. We grouped the ratings into three categories, where 1-2 is termed as Poor mental state, 3 is Average and 4-5 is Good. We have used a bar chart to show the relationship between time spent on devices and mental state. In Figure 6, as more time is spent on digital devices, we see that the number of respondents with Poor mental state increases in comparison to those with Good mental state.

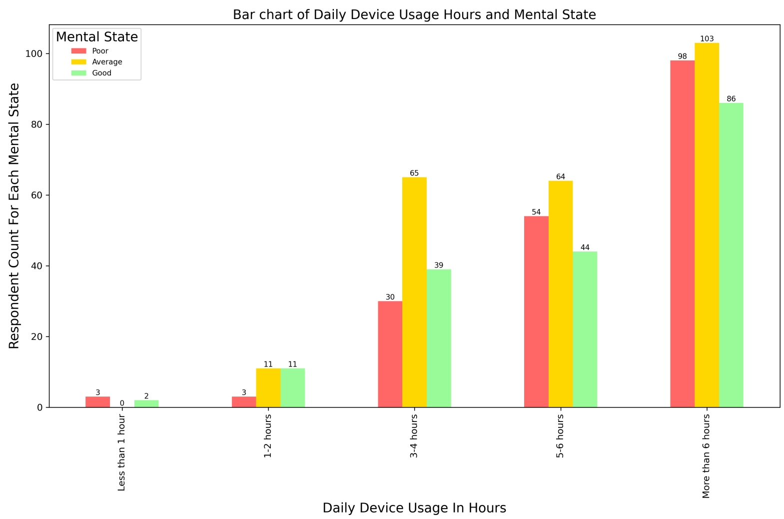


Figure 6: Bar chart of Daily Device Usage Hours and Mental State

3.4 Dataset Cleaning and preprocessing

At first, from the raw dataset, we carefully assessed each sample and decided to remove 11 entries, which resulted in **613** samples to be considered for our work. To load our dataset, we used Pandas. Since the column headers were in question format, we decided to rename each columns to improve readability. We then grouped mental well-being rating into three categories, which is summarized in table 3.

Table 3: Mental Well-being Rating Categories

Mental Well-being Rating	Category
1-2	0 (Poor Mental State)
3	1 (Average Mental State)
4-5	2 (Good Mental State)

To process the Age column of our dataset, we decided to extract the digits only from each entry. In our questionnaire, some of the questions allowed respondents to specify a different answer apart from the options given, and for this we grouped all the answers to **Others** category for all the columns.

To split our dataset into training and evaluation sets, we used the `train_test_split` function from the `scikit-learn` library. Our training set contained **521** samples, which is 85% of the entire dataset. Our evaluation set contained **92** samples, which is 15% of the entire dataset.

The Age feature of our dataset was scaled using a Min-max scaler. We used `MinMaxScaler` from `scikit-learn`, to learn the minimum and maximum values from the training set. The scaling transformation was applied using the learned minimum and maximum values on both the training and evaluation set. For every sample, Min-max scaling is applied according to equation 3:

$$X_{\text{new}} = \frac{X - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}} \tag{3}$$

To fill any null entries present in the Age feature, we calculated the median age of the training set and replaced all the null entries of both training and evaluation set using the median age of training set.

Finally, we separated the features and the target from the training and evaluation sets. We applied One-Hot Encoding to features without ordinal relationships among their categories, and Ordinal Encoding to the remaining features.

3.5 Algorithms

The classifiers used to train and evaluate our dataset are described in this section. We conducted three rounds of training and evaluation: first without any hyperparameter tuning and feature selection, next with feature selection applied, and finally, with both hyperparameter tuning and feature selection applied. However, we did not apply feature selection and hyperparameter tuning to all the algorithms we initially used. Details about hyperparameter tuning and feature selection is discussed in the next sub-section.

3.5.1 ZeroR classifier

ZeroR classifier is used to determine the baseline performance and hence sets as a benchmark for other classification methods. This classifier ignores all predictors and relies on the target only. It simply predicts the majority category.

3.5.2 Decision Tree classifier

The Decision Tree classifier [14] is a tree-like machine learning model, that recursively splits the dataset based on the most significant attributes. This recursion continues till data is split in such a way that it becomes homogeneous. In decision tree induction Iterative dichotomiser 3 (ID3) is used as the core algorithm [1]. To determine the best split during tree construction, entropy and information gain is used. To measure the impurity of an arbitrary node, we use entropy. For homogeneous attributes, entropy is zero, which means all instances belongs to the same class. Entropy increases as heterogeneity of instances increases. Entropy is given by formula 4:

$$E = -\sum_{i=1}^{n} p_i \log_2(p_i) \tag{4}$$

Here, n is the number of classes.

To select the next attribute, we use information gain.The attribute selected will have the highest information gain. To calculate information gain we use the formula 5:

$$\text{IG}(S, A) = \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} \cdot \text{Entropy}(S_v). \tag{5}$$

In Equation (5), A represents a known attribute, S_v is the subset of A, and v is the value of A.

3.5.3 Random Forest classifier

Random Forest [4] is an ensemble of Decision Trees. It builds multiple trees with different subsets of the data and features, and the final prediction is the mode of individual tree predictions. A prediction is given by formula 6:

$$\text{RF}(x) = \text{mode}\{H_1(x), H_2(x), \ldots, H_t(x)\} \tag{6}$$

Here $H_t(x)$ is the prediction of the t-th Decision Tree in the Random Forest.

3.5.4 Support Vector classifier

Support Vector Machines [3] can be used to solve classification problems. To categorize data points, it first maps the data to a high dimensional feature space. Then, it seeks to find the hyperplane that best separates the data points of different classes in the feature space.

3.5.5 Logistic Regression classifier

Logistic Regression [12] classifier can be used to solve multiclass classification problems. In our work, the target class has three categorical values, so the classifier is used for a multiclass classification task. Since Logistic Regression classifier is used to solve binary classification problems, the "one-vs-rest" strategy is used to handle multiclass problems. Here, a separate binary classifier is trained for each class, and during prediction the class is selected for which the corresponding classifier outputs the highest probability.In Equation (7), an input x belongs to class i if it maximizes $\max h_\theta^{(i)}(x)$ [1].

$$h_\theta(x) = g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}} \tag{7}$$

where,

$$g(z) = \frac{1}{1 + e^{-z}} \tag{8}$$

If the output of h_θ is less than 0.5, then the predicted output is 0, and if it is more than 0.5, then the predicted output is 1. Equation (8) is the Sigmoid function which maps real-valued numbers between 0 and 1.

3.5.6 K-Nearest Neighbour classifier

K-Nearest Neighbour [21] can be used to classify a data point based on the majority class among its k nearest neighbors in the feature space. To evaluate a test data point, the distance is measured between the test point and all other data points in the training set. Then the nearest k

data points are used to decide the class that the classifier predicts [13]. Equation 9 shows euclidean distance used to measure the distance.

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2} \tag{9}$$

3.5.7 Naïve Bayes classifier

Naïve Bayes classifier [15] uses Bayes' theorem. Provided prior knowledge, the probability of a hypothesis can be measured using equation 10.

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)} \tag{10}$$

$$P(x|y) = P(x_1|y) \cdot P(x_2|y) \cdot \ldots \cdot P(x_n|y) \tag{11}$$

Naïve Bayes makes an assumption that all input attributes are independent. In equation 12, X is the dependent vector and Y is the class we want to predict.

$$P(X_1, X_2, \ldots, X_n|Y) = \prod_{i=1}^{n} P(X_i|Y) \tag{12}$$

3.5.8 Boosting classifiers

In our work, we have used Gradient Boosting [7] and AdaBoost [6] as Boosting classifiers. Boosting refers to combination of several weak learners into a strong learner, making it an ensemble learner.

3.6 Feature Selection

We selected features using `SelectFromModel` available in the `scikit-learn` library. Feature selection is performed on the Decision Tree, Random Forest, Gradient Boosting, Logistic Regression, and AdaBoost models.

3.7 Hyperparameter Tuning

We used three different hyperparameter tuning techniques to find the best values of the classifiers' hyperparameters. We performed grid search using `GridSearchCV` and random search using `RandomizedSearchCV`. We have also used Optuna to optimize hyperparameters. In random search, each iteration tries a random combination of hyperparameter and records performance of the model. In grid search, the model is fit with each and every combination of hyperparameters possible. In each case, the best hyperparameters are selected. We performed hyperparameter tuning without feature selection on the Decision Tree, Random Forest, Support Vector, Gradient Boosting, Logistic Regression, AdaBoost, K-Nearest Neighbors, and Naive Bayes Classifier models. Additionally, we performed hyperparameter tuning with feature selection on the Decision Tree, Random Forest, Gradient Boosting, Logistic Regression, and AdaBoost models.

3.8 Explainable AI (XAI)

To implement explainable AI technique, we used LIME (Local Interpretable Model-agnostic Explanations) library to explain the predictions of a model. In our work, we used LIME to explain the prediction of Random Forest classifier without any feature selection or hyperparameter tuning.

4 Result analysis

This section discusses the result analysis of our work and gives a brief description of the evaluation metrics used to measure performance.

4.1 Evaluation Metrics

We used Accuracy, (Macro) Precision, (Macro) Recall and (Macro) F1-score to evaluate the performance of our models. The equations 13, 14, 15, 16 are used to calculated each of the metrics.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \tag{13}$$

$$\text{Macro Precision} = \frac{1}{N}\sum_{i=1}^{N}\frac{\text{True Positives}_i}{\text{True Positives}_i + \text{False Positives}_i} \tag{14}$$

$$\text{Macro Recall} = \frac{1}{N}\sum_{i=1}^{N}\frac{\text{True Positives}_i}{\text{True Positives}_i + \text{False Negatives}_i} \tag{15}$$

$$\text{Macro F1-score} = \frac{1}{N}\sum_{i=1}^{N}\frac{2 \times \text{Precision}_i \times \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \tag{16}$$

4.2 Performance evaluation

In table 4, performance evaluation of different classifier is tabulated with their train accuracy, test accuracy, precision, recall and F1 Score. The Decision Tree classifier, Random Forest classifier has a train accuracy of 100% which is the highest recorded. Gradient Boosting has a train accuracy of 96% which is the second highest. The ZeroR classifier recorded the lowest training accuracy of 40%. The highest evaluation accuracy was achieved by Support Vector classifier, which is 65%. Gradient Boosting, Logistic Regression classifier, K-Nearest Neighbour classifier and Naive Bayes recorded 63%, 60%, 60% and 59% respectively. Again, the lowest evaluation accuracy of 36% was achieved by the baseline classifier.

The highest precision value of 79% was achieved by the ZeroR classifier. Support Vector classifier and Gradient Boosting recorded precision values 68% and 67% respectively. AdaBoost achieved the lowest precision of 57%. The highest recall was achieved by Support Vector classifier, which was 54%. Gradient Boosting, Decision Tree, K-Nearest Neighbour achieved 62%, 60% and 60% recall values respectively. The lowest recall was recorded from the ZeroR classifier, with a value of 33%.

The highest F1-score was achieved by Support vector classifier with a score of 66%. Following this, Gradient Boosting achieved 63%, Logistic Regression achieved 60% and K-Nearest Neighbour achieved 60%. The Lowest F1-score was recorded from ZeroR classifier with a score of 18%. Overall, Support Vector classifier performed the best when compared to other classifiers.

Algorithms	**Train Accuracy (%)**	**Test Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1 score (%)**
ZeroR (baseline)	40	36	79	33	18
Decision Tree	100	60	60	60	59
Random Forest	100	57	60	57	57
SupportVector	**72**	**65**	**68**	**65**	**66**
Gradient Boosting	96	63	67	62	63
Logistic Regression	70	60	61	59	60
AdaBoost	62	55	57	56	56
K-Nearest Neighbour	69	60	60	60	60
Naive Bayes	63	59	62	58	59

Table 4: Performance metrics for different algorithms.

Algorithms	**Train Accuracy (%)**	**Test Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1 score (%)**
Decision Tree	100	59	60	59	59
RandomForest	**100**	**67**	**68**	**67**	**68**
Gradient Boosting	90	58	60	58	58
Logistic Regression	69	62	63	62	62
AdaBoost	62	55	57	56	56

Table 5: Performance metrics with feature selection applied for different algorithms.

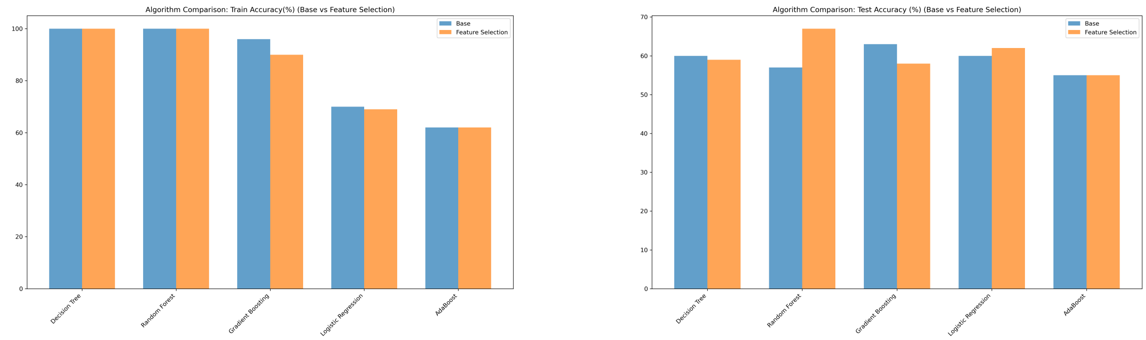
4.3 Performance evaluation with Feature Selection

Before hyperparameter optimization, feature selection was performed on some selected algorithms. The classifiers on which feature selection was performed are: Decision Tree, Random Forest, Gradient Boosting, Logistic Regression and AdaBoost. Table 5 summarizes the performance of the algorithms after feature selection. After feature selection, the train accuracy of Decision Tree classifier remained unchanged. The Test Accuracy slightly decreased from 60% to 59%. The precision remained unchanged as 60%. Recall slightly dropped from 60% to 59%. F1-score remained unchanged.

Random Forest performed significantly well. Its train accuracy remained uncharged, but the evaluation accuracy increased from 57% to 67%, precision increased from 60% to 68%, recall increased from 57% to 67% and F1-score increased from 57% to 68%.

The overall performance of Gradient Boosting dropped after feature selection, where the most significant drop was seen in evaluation accuracy from 63% to 58%. Other metrics dropped as well.

For Logistic regression classifier and AdaBoost, the metrics remained relatively same. After feature selection, Random Forest performed the best. Figure 7, 9, and 8 shows bar chart comparison of the results.



(a) Train Accuracy Comparison

(b) Test Accuracy Comparison

Figure 7: Train and Test Accuracy Comparison between Base evaluation and feature selected evaluation

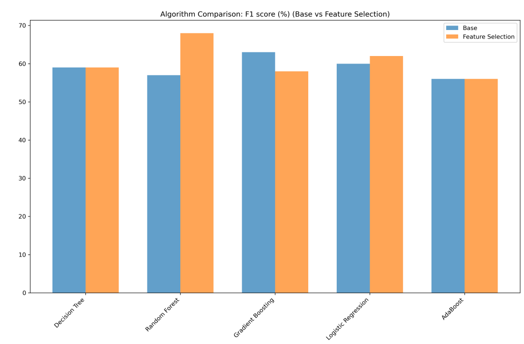
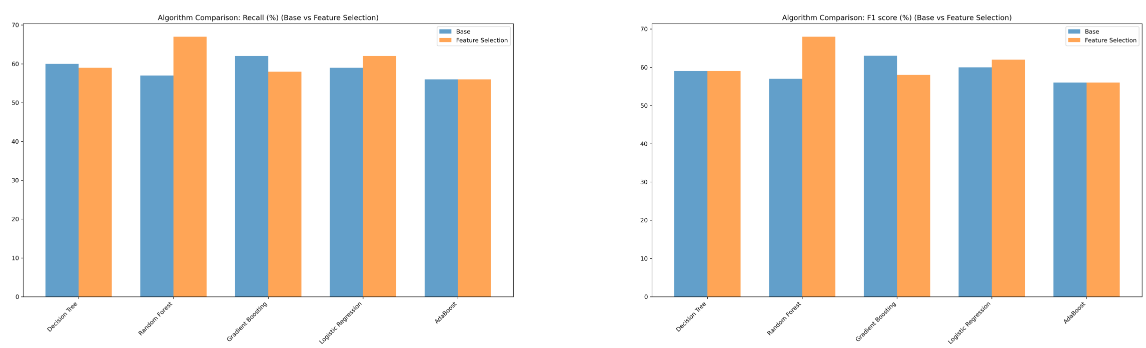


Figure 8: F1 Score Comparison between Base evaluation and feature selected evaluation



(a) Precision Comparison

(b) Recall Comparison

Figure 9: Precision and Recall Comparison between Base evaluation and feature selected evaluation

4.4 Performance evaluation with Hyperparameter Optimization

When our model was trained with three hyperparameter optimization techniques it was found that for Decision Tree the train accuracy for GridSearchCV and RandomizedSearchCV was 71% whereas for Optuna it was 68%. On the other hand, the test accuracy for each was 61%, 65% and 62% respectively which shows it performed better in test result but poorly in train set after hyperparameter optimization when compared to base evaluation.

For Random Forest it was seen that the train accuracy for GridSearchCV, RandomizedSearchCV and Optuna was 82%, 80% and 89% respectively. It also showed that for test accuracy it achieved 61%, 59% and 65%. Both train and test results after hyperparameter optimization varied greatly from the base evaluation results.

In case of Support Vector, Gradient Boosting and Logistic Regression each of their average results had similar accuracy as the base result in both train and test accuracy. In AdaBoost the result for train accuracy was similar to the base evaluation but performed better in the test set when compared to the base results.

K-Nearest Neighbour had train accuracy of 65% for GridSearchCV and 100% for the other two techniques, which showed a large difference than the base evaluation but performed worse in the train set than the base result.

Naive Bayes performed relatively the same in both train and test set compared to the base evaluation. Table 6 summarizes the results.

Algorithms	**Train Grid SearchCV (%)**	**Test Grid SearchCV (%)**	**Train Randomized SearchCV (%)**	**Test Randomized SearchCV (%)**	**Train Optuna (%)**	**Test Optuna (%)**
Decision Tree	71	61	71	65	68	62
Random Forest	82	61	80	59	89	65
Support Vector	73	61	81	65	70	65
Gradient Boosting	64	53	73	75	99	64
Logistic Regression	65	63	65	63	62	62
AdaBoost	64	63	64	63	63	63
K-Nearest Neighbour	65	51	100	50	100	57
Naive Bayes	63	60	63	59	64	59

Table 6: Accuracy scores with different hyperparameter tuning for various algorithms.

4.5 Performance evaluation with Hyperparameter Optimization and Feature Selection

Here both hyperparameter optimization and feature selection were applied. For Decision Tree the train accuracy for GridSearchCV, Randomized SearchCV and Optuna were 71%, 74% and 79% respectively. The outcome for the train set was much lower than the base result which was 100% and for the test set it showed similar results as the base. Random Forest had 76% and 64%, 81% and 63%, and 79% and 71% train accuracy and test accuracy for GridSearchCV, Randomized

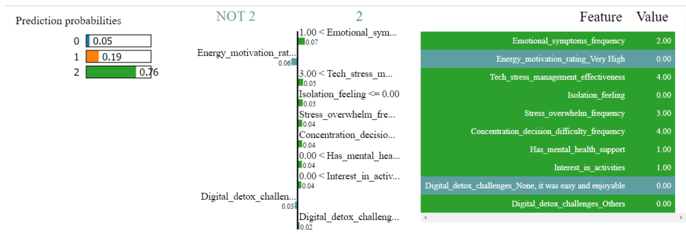
SearchCV and Optuna respectively. Showing poorer performance in the train set and slightly better in the test set. For Gradient Boosting the training set had a much lower accuracy value of 64% in GridSearchCV and Randomized SearchCV, and 73% in Optuna when compared to the base value of 96%. The test set had 53% accuracy for GridSearchCV and Randomized SearchCV which was less than 63% accuracy in the base result, and 69% for optuna which was slightly higher than that of the base result. Logistic regression's train accuracy was lower in all three techniques but had improved accuracy in the test set than that of the base evaluation. AdaBoost had an average of 62% accuracy in training set performance and 65% accuracy in the test set, showing better performance when compared to the base. Table 7 summarizes the results described above.

Algorithms	**Train Grid SearchCV (%)**	**Test Grid SearchCV (%)**	**Train Randomized SearchCV (%)**	**Test Randomized SearchCV (%)**	**Train Optuna (%)**	**Test Optuna (%)**
Decision Tree	71	62	74	63	79	70
Random Forest	76	64	81	63	79	71
Gradient Boosting	64	53	64	53	73	69
Logistic Regression	64	71	64	71	68	72
AdaBoost	62	65	62	65	61	69

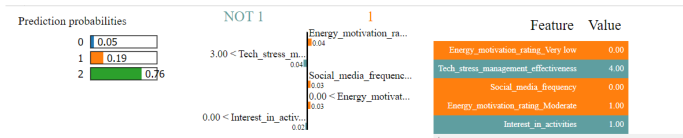
Table 7: Accuracy scores with different hyperparameter tuning and with feature selection for various algorithms.

4.6 Results from LIME explainer

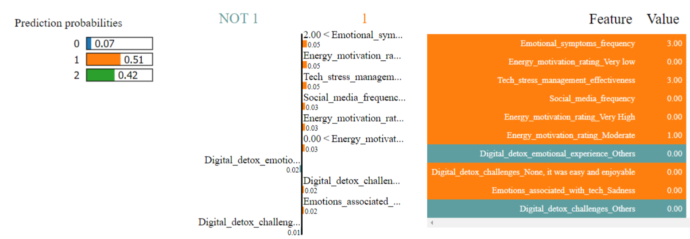
Figure 10 shows our results from LIME explainer. In figure 10a, for the first instance in the test set, the Random Forest Model predicted class 2 (Good Mental State) with a confidence of 0.76. Figure 10c shows that explainer predicts class Poor Mental state with a confidence of 0.19, with 5 features, and out of which 3 contributes towards this prediction. Figure 10d shows the local explanation of the first class (Poor Mental State).



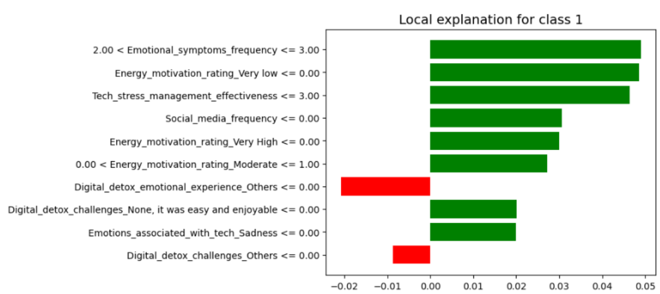
(a) Explanation of the first instance in the test set for class 2.



(b) Explanation of the first instance in the test set with 5 features.



(c) Explanation of the 10th instance in the test set.



(d) Local explanation for class 1 (poor mental state).

Figure 10: Results from Lime Explainer

5 Discussion

In this work, we used Machine Learning classifiers to predict the mental state of individuals by different factors associated with technology usage. Unlike other works, we incorporated questions based on the ability of an individual to practice digital detox, which means to restrain oneself from technology usage. We also asked questions on sleep quality, and other coping mechanisms which could affect mental state. Given all the features, we decided to predict the mental state using three target class: Poor mental state, average mental state and Good Mental state.

After the initial survey, we selected 613 best suited entries to train our classifiers. Given the size of the data, and the number of features, our model performed adequately. A maximum evaluation accuracy of 72% was recorded from all our models after optimization.

The entire training and testing was done on the following desktop configuration: CPU: Ryzen 5 3600 @ 3.6GHz, 16GB Ram @3200 Bus speed.System Type: 64-bit operating system, x64-based processor

6 Limitation of the work

Despite implementing different feature selection techniques and hyperparameters, there were some constraints in our project. Firstly, since we collected data to create our own dataset, the dataset was not large enough to provide the most optimal result. The dataset mostly had participants of age group 22 and 23 which may have caused the model to learn the mental state of University students . Due to the limitation of data the resulting outputs had average accuracy value. The model used a few hyperparameter tuning and feature selection techniques. Lastly, no deep learning model like Artificial Neural Network(ANN) or Convolutional Neural Network(CNN) were used which prevented the model from showing its performance in other categories of models.

7 Conclusion and future scope

Our project aimed to determine the impact of technology on mental health. To execute the project we collected data that consisted of 624 responses out of which 613 was used.After thorough preprocessing of the data we did feature selection.The mental health was rated in three classes - poor, neutral and good. The model was carried out using nine classifiers, one feature selection and three hyperparameter tuning techniques. The maximum achieved accuracy was 72% for logistic regression, and recall was 67% for Random Forest. Lastly, LIME explainer was used to get a deeper understanding of how the model derived its predictions.

The project has scope of performing better with a greater number of data and by implementing more different kinds of hyperparameter and feature selection techniques. The project could be made to perform better by using other models such as CNN and ANN.

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